Yi-Chi (Trista) Chen

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Education

San José State University — San Jose, CA

B.S. in Software Engineering — *Expected May 2027*

Relevant Coursework: Data Structures & Algorithms, Computer Organization & Architecture, Assembly Language, Operating Systems, Database Systems, Machine Learning for Big Data

Experience

Firmware Software Engineer Intern — April 2026 – May 2026

Alef — *San Mateo, CA*

- Designed ESP32-based wheel calibration system using dual Hall sensors, achieving **sub-millimeter** zero-position recovery after power loss
- Engineered CAN (TWAI) motor control pipeline supporting velocity and incremental positioning at **100 Hz** loop speed
- Developed automated homing algorithm with baseline hysteresis, reducing signal-noise false triggers by **40%** during high-EMI conditions
- Exposed diagnostic web interface for real-time motor calibration, decreasing field testing setup time by **25%**

Software Engineer – Mission Control Team — Jan 2026 – Aug 2026

SJSU Robotics Club — *San Jose, CA*

- Integrated React frontend and Python backend over CAN protocol, reducing end-to-end teleoperation latency by **35% (<50ms round-trip)**
- Engineered command/response parser with ACK handling and auto reconnect, lowering packet loss to **<0.5%** over lossy links
- Built dual slider/gamepad UI with configurable joint limits and deadzones, improving operator command accuracy across **100%** of test trials
- Implemented real-time telemetry rendering (commanded vs. feedback TX indicators) running at **60 FPS** for operator situational awareness

Software Engineer Intern — Mar 2026 – May 2026

SJSU Software and Computer Engineering Society — *San Jose, CA*

- Architected **SCEvents** using Go, Kafka, Redis, and MongoDB, supporting **1,000+** concurrent users with zero downtime
- Engineered Kafka event-driven pipeline to process registration/waitlist streams, handling **500+ msg/sec** throughput
- Implemented Redis distributed locking and caching, eliminating race conditions under k6 load tests of **10,000+** write spikes

Associate Test Technician — June 2026 – Present

Element Materials Technology — *Morgan Hill, CA*

- Executed EMC/RF engineering validation across Wi-Fi 6E, Bluetooth, and UWB technologies using multi-axis antenna arrays
- Analyzed peak/average RF measurements and tuned transmit power/EIRP, accelerating regulatory compliance certification timelines
- Isolated RF measurement anomalies across frequency bands and configurations, increasing test repeatability across **50+** device suites

Machine Learning Engineer — Feb 2026 – May 2026

SJSU AI/ML Club (Industry-Led Project, Nuvoton) — *San Jose, CA*

- Trained YOLOv8 models for real-time headcount estimation, achieving **mAP@0.5 of 0.979** and **mAP@0.5:0.95 of 0.651**
- Executed 200+ hyperparameter experiments to optimize precision-recall trade-offs, improving overall accuracy by **>30%**
- Deployed optimized models on edge hardware (Ethos-U55), maintaining real-time inference at **20–22 FPS**

Projects

Air Quality Monitoring System | *Arduino, C++, Sensors* — April 2025 – June 2025

- Engineered embedded monitoring system on Arduino UNO using gas sensors, processing environmental readings at **10 Hz**
- Implemented threshold control logic to trigger fan activation and LED alerts within **<200ms** of detecting unsafe gas spikes
- Optimized hardware/software communication via system testing, achieving **99.5%** uptime across **72-hour** stress tests

Technical Skills

Languages: C, C++, Python, Go, Java, JavaScript, TypeScript, Assembly, SQL
Embedded & Firmware: ESP32, CAN (TWAI), UART, USB, Ethos-U55, Motor Control, Hall Sensors, Hardware Debugging
Systems & Frameworks: React, FastAPI, PyTorch, Socket.IO, Kafka, Redis, MongoDB, Git, Docker, Linux,